

Hydraulically-Informed Graph Learning for Monitoring of Sewer Networks under Data Sparsity

Abstract

Urban sewer networks are difficult and expensive to monitor comprehensively, as installing sensors at every manhole or pipe section is infeasible. Nevertheless, utilities must still make timely decisions regarding inspections, maintenance, and operational responses.

This paper presents a hydraulically-informed graph learning framework that treats sewer networks as connected systems and, therefore, infers asset conditions in monitored and unmonitored sections. Aiming for greater representation accuracy, the graph-based modelling considers structural and hydraulic relationships across the sewer network rather than assets in isolation. Consequently, when a sensor detects flow changes, anomalies, or early warning signals, the graph-based model maps the likelihood of the issue across the broader infrastructure. This informs the decision-maker of the asset condition and health, regardless of whether the original issue is close to a monitored point.

The result is system-wide situational awareness from limited data. Utilities can identify likely problem areas, prioritise inspections, and act before failures occur. This approach provides a cost-effective upgrade to existing monitoring infrastructure without the need for additional hardware.

Links

<https://udam.home.blog/euro-sam/>

<https://form.jotform.com/260474430626354>

<https://form.jotform.co/udam/join>

